

# Commercial Development Proposal & Statement of Work

**Target Project Profile:** Nexus Ludo Clone (Real-Money Gaming Platform)  
**Scope of Analysis:** Core Architecture Implementation & Milestone Quotation

## 1. Fixed Development Fee & Core Deliverables Matrix

This section defines the absolute, fixed engineering cost for programming and launching the full-stack system layout according to the high-concurrency production blueprint. All structural code customizations, wallet rules, and system logic setups are fully covered under this execution budget.

|                                 |                 |                                            |  |  |  |
|---------------------------------|-----------------|--------------------------------------------|--|--|--|
| Contract Phase                  | Fixed Fee (INR) | Engineering Scope & Deliverables Checklist |  |  |  |
| Full-Stack Platform Engineering |                 | ₹55,000                                    |  |  |  |
| (Fixed Contract)                |                 |                                            |  |  |  |

- Cross-Platform Mobile Client:** Compiling the React Native + Expo bare workflow into a native Android application asset with lightweight, fluid 60 FPS vector canvas layouts.
- Authoritative Game Server:** Deploying an asynchronous Node.js + Express backend running a strict central grid coordinate system to fully control dice outcomes and match referee operations.
- Real-Time Turn Manager:** Coding the active 15-second individual player turn countdown alongside automated 60-second connection failure grace windows to prevent room stalling.
- ACID Transaction Ledger:** Building a double-entry database architecture via Mongoose where balances are safely computed at runtime from raw transactional rows to block double-spending.
- Anti-Cheat Shielding:** Installing middleware filters protecting input queries against NoSQL injection vectors, along with Redis-based API token blacklists and individual IP network rate limiters.

Total Engineering Contract Outlay: ₹55,000 INR (Fixed Development Fee Only)

## 2. Granular Milestone & Cost Breakdown

To ensure total transparency, the fixed development fee is distributed across five logical subsystems based on specialized technical complexity and implementation depth:

|                                |                |                                             |                                                  |  |  |
|--------------------------------|----------------|---------------------------------------------|--------------------------------------------------|--|--|
| Technical Subsystem            | Milestone Cost | Granular Scope Details & Feature Inclusions |                                                  |  |  |
| 1. Frontend Client Integration |                | ₹15,000                                     | Configuring strict TypeScript type-safety across |  |  |

screens; styling user wallet layouts, cash loading portals, deposit panels, and writing background OTA update loops via Expo Application Services. | | **2. Authoritative Match Logic** | ₹15,000 | Writing server-side game mechanics: generating random integers (1-6), validating token placement locks, handling pawn combat events, and mapping the absolute victory index (Index 56). | | **3. Gateway & Compliance Engines** | ₹11,000 | Writing secure deep-link intent scripts for mobile UPI handshakes (GPay/PhonePe); writing SHA-256 HMAC cryptographic verification hashes; establishing automatic 30% Net TDS tax calculation tables. | | **4. Redis Memory Cache Stratum** | ₹7,000 | Setting up high-speed memory pools for fluid game sessions; coding the automated 20-second matchmaker grid queues; designing weighted simulated bot scripts to keep match lobbies filled. | | **5. Cluster DevOps & Hardening** | ₹7,000 | Configuring multi-core Node.js process managers via PM2 Cluster Mode; linking clustered threads together over Redis Pub/Sub adapters; installing restrictive CORS policies and Helmet security headers. | | **Total** | ₹55,000 | |

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### 3. External Client-Owned Infrastructure Responsibilities (Billed Separately)

To prevent budget creep and maintain operational safety, the digital structural assets detailed below must be owned directly by the client. These are procured externally and are not deducted from the development fee:

| Structural Necessity | Procurement Model | App Functionality Justification | | :--- | :--- | :--- | | **High-Compute Virtual Host** | Monthly / Annual Account | Dedicated infrastructure (AWS / DigitalOcean) optimized to sustain ongoing WebSocket communication lines across simultaneous matches. | | **Official Domain Identity** | Annual Registration | A secure custom URL (.com / .in) required to manage web API routing layers and store the downloadable Android mobile build. | | **Transactional SMS Gateway** | Usage-Based (Per SMS Packet) | Third-party SMS utility (Twilio / MSG91) providing instant mobile verification OTP blocks to secure authorization flows and eliminate bot registration. | | **Payment Gateway Merchant Profile** | Transaction Commission Cut | Commercial corporate merchant accounts (Cashfree / RazorpayX) to execute direct consumer payins and process immediate automated bank payouts. |